

Supplementary Material

Pathology-Knowledge-Guided Structured Annotation for Clinical Speech Audio

This supplementary material provides additional analyses, implementation details, and reproducibility information for the main paper. It includes population- and label-level support statistics, validation and sensitivity analyses for the weak-label pipeline, uncertainty analyses, serialization-error analysis, model configurations, complete Qwen3-Omni prompts and demonstration sets, audio preprocessing details, and JSON parsing and repair procedures.

1 Population-wise support of the 22-label schema

The 150 expert-reference records contain 50 recordings from each clinical population (30 training, 10 validation, and 10 test records per population). Table 1 reports positive record-level support for each label. Table 2 summarizes dimension coverage using both positive label instances and record-level presence.

Table 1: Expert-reference label support across the three clinical populations. Each population contains 50 annotated records.

Dimension	Label	Aphasia	Depression	Dysarthria	Total
Respiration	Onset difficulty	24	0	3	27
Respiration	Phonation interruption	23	3	3	29
Respiration	Audible inhalation	1	0	1	2
Phonation	Hoarse/rough voice	12	4	33	49
Phonation	Breathiness	4	1	8	13
Phonation	Vocal tremor	0	1	11	12
Phonation	Pressed voice	0	6	8	14
Phonation	Weak voice	2	10	16	28
Articulation	Imprecise consonants	23	1	47	71
Articulation	Reduced vowel fullness	23	0	45	68
Articulation	Hypernasality	0	0	18	18
Articulation	Hyponasality	0	0	1	1
Prosody	Abnormal stress	6	0	16	22
Prosody	Unnatural rhythm	28	2	47	77
Prosody	Word-by-word speech	11	0	38	49
Prosody	Fast speech	6	0	0	6
Prosody	Slow speech	13	1	33	47
Prosody	Monotone prosody	0	0	22	22
Prosody	Excessive fillers	0	2	0	2
Additional events	Sighing	0	0	0	0
Additional events	Crying-like voice	0	1	0	1
Additional events	Laughter	0	1	0	1

Table 2: Dimension-level support across clinical populations. Label-instance support sums positive labels within a dimension; record presence counts records containing at least one positive label in that dimension.

Dimension	Label instances			Record presence		
	Aphasia	Depression	Dysarthria	Aphasia	Depression	Dysarthria
Respiration	48	3	7	31	3	5
Phonation	18	22	76	13	17	36
Articulation	46	1	111	30	1	48
Prosody	64	5	156	31	4	48
Additional events	0	2	0	0	2	0

2 Held-out validation of the frozen weak-label pipeline

The frozen weak-label pipeline was applied without further adjustment to the 30 expert-annotated validation records. It achieved a Micro-Precision of 0.4609, Micro-Recall of 0.5048, and Micro-F1 of 0.4818. Tables 3 and 4 provide dimension- and label-level agreement. Undefined quantities caused by zero denominators are shown as “–”.

Table 3: Dimension-wise agreement of frozen weak labels with expert references on the 30-record validation split. Support denotes positive reference label instances.

Dimension	Support	Precision	Recall	F1
Respiration	8	0.1154	0.3750	0.1765
Phonation	21	0.1429	0.0476	0.0714
Articulation	28	0.6970	0.8214	0.7541
Prosody	48	0.5306	0.5417	0.5361
Additional events	0	–	–	–

Table 4: Label-wise agreement of frozen weak labels with expert references on the 30-record validation split.

Dimension	Label	Ref. support	Pred. +	Precision	Recall	F1	κ
Respiration	Onset difficulty	4	13	0.1538	0.5000	0.2353	0.0394
Respiration	Phonation interruption	3	13	0.0769	0.3333	0.1250	-0.0448
Respiration	Audible inhalation	1	0	–	0.0000	0.0000	0.0000
Phonation	Hoarse/rough voice	7	0	–	0.0000	0.0000	0.0000
Phonation	Breathiness	3	0	–	0.0000	0.0000	0.0000
Phonation	Vocal tremor	4	5	0.0000	0.0000	0.0000	-0.1739
Phonation	Pressed voice	3	1	0.0000	0.0000	0.0000	-0.0526
Phonation	Weak voice	4	1	1.0000	0.2500	0.4000	0.3662
Articulation	Imprecise consonants	14	13	0.7692	0.7143	0.7407	0.5291
Articulation	Reduced vowel fullness	11	13	0.7692	0.9091	0.8333	0.7235
Articulation	Hypernasality	3	7	0.4286	1.0000	0.6000	0.5349
Articulation	Hyponasality	0	0	–	–	–	–
Prosody	Abnormal stress	5	6	0.1667	0.2000	0.1818	0.0000
Prosody	Unnatural rhythm	14	12	0.6667	0.5714	0.6154	0.3243
Prosody	Word-by-word speech	11	8	0.7500	0.5455	0.6316	0.4670
Prosody	Fast speech	2	0	–	0.0000	0.0000	0.0000
Prosody	Slow speech	12	14	0.5000	0.5833	0.5385	0.1892
Prosody	Monotone prosody	4	7	0.5714	1.0000	0.7273	0.6715
Prosody	Excessive fillers	0	2	0.0000	–	0.0000	0.0000
Additional events	Sighing	0	0	–	–	–	–
Additional events	Crying-like voice	0	0	–	–	–	–
Additional events	Laughter	0	0	–	–	–	–

3 Threshold sensitivity of the frozen weak-label rules

A one-at-a-time sensitivity analysis perturbed each analyzable continuous threshold to $0.8\times$, $0.9\times$, $1.0\times$, $1.1\times$, and $1.2\times$ its frozen value while keeping all other rules fixed. The $1.0\times$ condition exactly reproduced the original 30-record validation predictions (Micro-F1 = 0.4818). Table 5 reports the resulting Micro-F1 ranges.

Table 5: One-at-a-time threshold sensitivity on the 30-record validation split. Tested range denotes the minimum and maximum numerical threshold values across the $0.8\times$ – $1.2\times$ perturbations.

Threshold	Affected label	Baseline	Tested range	Micro-F1 range
<code>hard_attack_ratio_gt</code>	Onset difficulty	0.18	0.144–0.216	0.4589–0.4818
<code>break_frequency_gt</code>	Phonation interruption	0.2	0.16–0.24	0.4753–0.4840
<code>break_ratio_gt</code>	Phonation interruption	0.05	0.04–0.06	0.4818–0.4818

Threshold	Affected label	Baseline	Tested range	Micro-F1 range
ast_breath_probability_gt	Audible inhalation	0.15	0.12–0.18	0.4818–0.4818
hoarse_cppts_lt	Hoarse/rough voice	0.12	0.096–0.144	0.4775–0.4818
hoarse_hnr_lt	Hoarse/rough voice	4	3.2–4.8	0.4818–0.4818
hoarse_shimmer_ge	Hoarse/rough voice	1.1	0.88–1.32	0.4818–0.4818
breathy_cppts_lt	Breathiness	0.1	0.08–0.12	0.4818–0.4818
breathy_hnr_lt	Breathiness	6	4.8–7.2	0.4818–0.4818
tremor_jitter_ge	Vocal tremor	0.05	0.04–0.06	0.4732–0.4885
tremor_shimmer_ge	Vocal tremor	1.1	0.88–1.32	0.4818–0.4818
squeezed_loudness_ge	Pressed voice	0.5	0.4–0.6	0.4818–0.4818
squeezed_cppts_ge	Pressed voice	0.18	0.144–0.216	0.4818–0.4818
squeezed_hnr_ge	Pressed voice	9	7.2–10.8	0.4818–0.4840
weak_loudness_lt	Weak voice	0.45	0.36–0.54	0.4749–0.4818
weak_cppts_lt	Weak voice	0.12	0.096–0.144	0.4818–0.4818
weak_hnr_lt	Weak voice	5	4–6	0.4818–0.4818
fast_rate_gt	Fast speech	3.3	2.64–3.96	0.4615–0.4818
slow_rate_lt	Slow speech	2.55	2.04–3.06	0.4476–0.4828
syllabic_rate_lt	Word-by-word speech	2.8	2.24–3.36	0.4135–0.4955
syllabic_interval_cv_lt	Word-by-word speech	0.45	0.36–0.54	0.4135–0.4867
unnatural_interval_std_gt	Unnatural rhythm	290	232–348	0.4779–0.4862
monotone_f0_cv_lt	Monotone prosody	0.13	0.104–0.156	0.4775–0.4818
monotone_loudness_cv_lt	Monotone prosody	0.85	0.68–1.02	0.4537–0.4818
stress_ratio_gt	Abnormal stress	8	6.4–9.6	0.4629–0.4818
filler_rate_gt	Excessive fillers	25	20–30	0.4775–0.4818
filler_ratio_gt	Excessive fillers	0.135	0.108–0.162	0.4775–0.4818
panns_laugh_probability_gt	Laughter	0.15	0.12–0.18	0.4818–0.4818
panns_cry_probability_gt	Crying-like voice	0.15	0.12–0.18	0.4818–0.4818
panns_sigh_probability_gt	Sighing	0.15	0.12–0.18	0.4818–0.4818
vowel_centroid_std_lt	Reduced vowel fullness	900	720–1080	0.4507–0.4818
hypernasal_nor_lower_gt	Hypernasality	1.5	1.2–1.8	0.4796–0.4818
hypernasal_hfr_mean_lt	Hypernasality	0.05	0.04–0.06	0.4818–0.4818
hyponasal_nor_upper_lt	Hyponasality	1	0.8–1.2	0.4818–0.4818

4 Test-set label support

Table 6 reports positive expert-reference support for all 22 labels on the 30-record test split, separated by clinical population. Each population contributes 10 test records. The three additional-event labels have zero positive test support and therefore do not receive dimension-level precision, recall, or F1 in the main paper.

Table 6: Expert-reference label support on the 30-record test split by clinical population.

Dimension	Label	Aphasia	Depression	Dysarthria	Total
Respiration	Onset difficulty	6	0	1	7
Respiration	Phonation interruption	6	0	1	7
Respiration	Audible inhalation	0	0	0	0
Phonation	Hoarse/rough voice	1	1	6	8
Phonation	Breathiness	0	0	2	2
Phonation	Vocal tremor	0	0	3	3
Phonation	Pressed voice	0	0	3	3
Phonation	Weak voice	0	0	5	5
Articulation	Imprecise consonants	7	0	10	17
Articulation	Reduced vowel fullness	7	0	10	17
Articulation	Hypernasality	0	0	3	3
Articulation	Hyponasality	0	0	1	1
Prosody	Abnormal stress	1	0	3	4

Dimension	Label	Aphasia	Depression	Dysarthria	Total
Prosody	Unnatural rhythm	8	0	10	18
Prosody	Word-by-word speech	2	0	9	11
Prosody	Fast speech	2	0	0	2
Prosody	Slow speech	2	0	7	9
Prosody	Monotone prosody	0	0	7	7
Prosody	Excessive fillers	0	0	0	0
Additional events	Sighing	0	0	0	0
Additional events	Crying-like voice	0	0	0	0
Additional events	Laughter	0	0	0	0

Table 7: Dimension-level support on the 30-record test split.

Dimension	Label-instance support	Record-presence support
Respiration	14	9
Phonation	21	10
Articulation	38	18
Prosody	51	18
Additional events	0	0

5 Complete weak-label generation rules

Table 8 lists the final deterministic rules used by the frozen weak-label pipeline. The thresholds shown are the final post-calibration values, including substitutions made by the frozen `threshold_final` wrapper.

Feature and model configuration. The general acoustic pipeline loads 16-kHz audio. Voice features are computed after `librosa.effects.split` with `top_db=25`, frame length 2048, and hop length 512; a retained-voice proportion below 0.35 sets the voice-quality flag to low quality. `openSMILE` uses eGeMAPSv02 LLD/Functionals. Articulation features use `trim(top_db=20)` and the STFT bands specified in Table 8. PANNs uses Cnn14 at 32 kHz with window 1024, hop 320, 64 mel bins, 50–14,000 Hz frequency limits, 527 output classes, and checkpoint `Cnn14_mAP=0.431.pth`. Silero VAD is loaded from the project’s local `silero-vad` copy.

Implementation-specific identifiers. The frozen source records PANNs event outputs by indices 16, 22, and 26. AST breathing-event classes are selected dynamically from the model configuration using the keyword set shown in Table 8.

Table 8: Complete deterministic weak-label rules used by the frozen pipeline. Inclusive inequalities follow the implementation.

Dimension	Label	Evidence / feature	Frozen decision rule	Aggregation / implementation
Respiration	Onset difficulty	RMS energy, onset detection, onset-window gradient	For each onset, count a hard onset when the energy gradient over the following 10 frames exceeds 0.012; assign the label when hard-onset ratio > 0.18 .	RMS frame length 256, hop 128; only onset windows with at least two frames enter the ratio.
Respiration	Phonation interruption	Silero VAD segments, <code>pyin</code> voiced flag, gap duration	Retain 30–200 ms gaps with local voiced-context mean > 0.3 ; assign when break frequency > 0.2 or break-duration ratio > 0.05 .	Silero uses minimum silence 30 ms and minimum speech 50 ms. The <code>pyin</code> context is evaluated around the gap midpoint (± 5 frames).
Respiration	Audible inhalation	AST sigmoid probabilities and <code>id2label</code>	Select AST classes whose names contain breath , gasp , pant , wheeze , or sniff ; assign when the maximum selected probability > 0.15 .	Whole waveform is passed to AST; aggregation is the maximum over matching output classes.
Phonation	Hoarse/rough voice	CPPS p80, HNR p20, shimmer p80	$CPPS \leq 0.12$ and $(HNR \leq 4.0 \text{ or } shimmer \geq 1.10)$, with voice-quality flag ok .	Features are aggregated per audio after voice-region cropping; CPPS is a custom cepstral series.
Phonation	Breathiness	CPPS p80, HNR p20	$CPPS \leq 0.10$ and $HNR \leq 6.0$, with voice-quality flag ok .	Same voice-feature aggregation as above.
Phonation	Vocal tremor	Jitter p80, shimmer p80	$Jitter \geq 0.05$ and $shimmer \geq 1.10$, with voice-quality flag ok .	Same voice-feature aggregation as above.
Phonation	Pressed voice	Loudness p20, CPPS p80, HNR p20	$Loudness \geq 0.5$, $CPPS \geq 0.18$, and $HNR \geq 9.0$, with voice-quality flag ok .	Same voice-feature aggregation as above.
Phonation	Weak voice	Loudness p20, CPPS p80, HNR p20	$Loudness \leq 0.45$ and $(CPPS \leq 0.12 \text{ or } HNR \leq 5.0)$, with voice-quality flag ok .	Same voice-feature aggregation as above; 0.45 is the final wrapper value.
Articulation	Imprecise consonants	ZCR p95, spectral-flux standard deviation	Assign when ZCR p95 < 0.35 or spectral-flux SD < 1.0 .	Computed on the trimmed 16-kHz waveform; 0.35 is the final wrapper value.
Articulation	Reduced vowel fullness	Spectral-centroid standard deviation	Assign when centroid SD < 900 .	Standard deviation across spectral-centroid frames; 900 is the final wrapper value.
Articulation	Hypernasality	Active-frame nasal/oral ratio (NOR) p20 and high-frequency ratio mean	$NOR\ p20 > 1.5$ and mean high-frequency ratio < 0.05 .	STFT power bands: 200–500 Hz nasal, 500–2500 Hz oral, and > 2500 Hz high-frequency; active frames have RMS above p20.
Articulation	Hyponasality	Active-frame NOR p95	$NOR\ p95 < 1.0$.	Same STFT bands and active-frame selection; at least 10 valid active frames are required by the feature routine.
Prosody	Abnormal stress	Loudness p80/p20 ratio	Assign when $p80/(p20 + 10^{-6}) > 8.0$.	Loudness percentiles are read from the openSMILE Functionals output.
Prosody	Unnatural rhythm	RMS-peak interval SD, syllabic rate, interval CV	Interval SD > 290 ms, or (syllabic rate < 2.8 and interval CV < 0.45).	RMS frame length 512, hop 256; peak height = $\max(RMS_{dB}) - 25$ dB, distance 10, prominence 3.
Prosody	Word-by-word speech	Syllabic rate, interval CV	Syllabic rate < 2.8 and interval CV < 0.45 .	RMS peaks define event times; rate is peak count divided by first-to-last peak duration; CV is interval SD divided by mean interval.
Prosody	Fast speech	Syllabic rate	Syllabic rate > 3.3 .	Same RMS-peak rate; no rhythm labels are returned when fewer than two peaks are detected.
Prosody	Slow speech	Syllabic rate	Syllabic rate < 2.55 .	Same RMS-peak rate; 2.55 is the final wrapper value.
Prosody	Monotone prosody	F0 normalized SD, loudness normalized SD	F0 normalized SD < 0.13 and loudness normalized SD < 0.85 .	Values are read from openSMILE Functionals; 0.13 and 0.85 are final wrapper values.
Prosody	Excessive fillers	SenseVoiceSmall decoded text, fillers/minute, filler ratio	Assign when fillers/minute > 25 or filler ratio > 0.135 .	After removing special tokens, count the fixed Mandarin filler lexicon {嗯, 啊, 呃, 哎, 那个, 这个, 就是, 然后, 就}; rate uses audio duration and ratio uses cleaned character count.
Additional events	Sighing	PANNs Cnn14 <code>clipwise_output</code> [26]	Assign when probability > 0.15 .	One full-audio Cnn14 forward pass; no additional temporal pooling.
Additional events	Crying-like voice	PANNs Cnn14 <code>clipwise_output</code> [22]	Assign when probability > 0.15 .	One full-audio Cnn14 forward pass; no additional temporal pooling.
Additional events	Laughter	PANNs Cnn14 <code>clipwise_output</code> [16]	Assign when probability > 0.15 .	One full-audio Cnn14 forward pass; no additional temporal pooling.

6 Bootstrap uncertainty analyses

6.1 S4 versus S0

We quantified uncertainty in the difference between the S4 label-aware construction and the S0 human-only baseline using 10,000 paired record-level bootstrap resamples of the common 30-record test set. The same resampled record indices were used for both systems, and training seeds were matched across S0 and S4. Differences are defined as S4 minus S0.

Table 9: Paired record-level bootstrap comparison between S4 and S0. Positive Hamming-loss differences indicate higher loss for S4.

Metric	Observed Δ	Bootstrap mean Δ	95% CI	$\Delta > 0$	$\Delta < 0$	$\Delta = 0$
Micro-F1	0.0052	0.0046	[-0.0039, 0.0150]	0.9661	0.0125	0.0214
Exact Match	0.0334	0.0333	[-0.0031, 0.0780]	0.9125	0.0875	0.0000
Hamming Loss	0.0035	0.0036	[-0.0019, 0.0117]	0.8813	0.0442	0.0745

6.2 SCSA versus Qwen3-Omni

Cross-system uncertainty was estimated with 10,000 common-record bootstrap resamples of the same 30-record test set. Within each resample, metrics were computed for all five SCSA training runs and all five Qwen demonstration sets and then averaged within system. Differences are defined as SCSA minus Qwen3-Omni.

Table 10: Common-record bootstrap comparison between SCSA S4 and Qwen3-Omni Q2.

Metric	Observed Δ	Bootstrap mean Δ	95% CI	$\Delta > 0$	$\Delta < 0$	$\Delta = 0$
Precision	0.3058	0.3289	[0.2047, 0.4599]	1.0000	0.0000	0.0000
Recall	-0.2378	-0.2637	[-0.3953, -0.1276]	0.0002	0.9998	0.0000
Micro-F1	0.0393	0.0468	[0.0176, 0.0760]	0.9504	0.0493	0.0003

7 Flat-22 label-level error analysis

The SCSA serialization ablation was further examined at label level over the five matched training seeds. Differences below are defined as Flat-22 minus five-dimensional serialization. The largest recall gain occurred for weak voice ($\Delta R = +0.8000$), while monotone prosody showed the largest F1 gain ($\Delta F1 = +0.7692$). The largest precision decrease occurred for slow speech ($\Delta P = -0.3571$). Weak voice also produced the largest increase in false positives (+10), whereas hoarse/rough voice showed the largest reduction in false negatives (-6).

Table 11: Largest label-level changes from five-dimensional to Flat-22 serialization.

Change	Label	Difference
Largest recall increase	Weak voice	+0.8000
Largest F1 increase	Monotone prosody	+0.7692
Largest precision decrease	Slow speech	-0.3571
Largest FP increase	Weak voice	+10
Largest FN decrease	Hoarse/rough voice	-6

Across the 150 record-seed cases, the five-dimensional and Flat-22 representations yielded 49 and 6 exclusive exact matches, respectively, with no case in which both serializations were exactly correct. The corresponding exact-status overlap is shown in Table 12. The most frequent failure combinations were phonation+articulation+prosody and respiration+articulation+prosody for the five-dimensional representation (five cases each), whereas phonation-only errors were most frequent for Flat-22 (11 cases). Among cases correctly matched only by the five-dimensional representation, weak voice was the most frequent additional false-positive label under Flat-22; no omitted false-negative label was observed in these cases.

Table 12: Exact-match overlap between the two SCSA serialization variants across 150 record-seed cases.

	Flat-22 correct	Flat-22 incorrect
Five-dimensional correct	0	49
Five-dimensional incorrect	6	95

8 SCSA architecture and optimization configuration

Table 13 summarizes the architecture and training configuration used for SCSA.

Table 13: SCSA architecture and optimization configuration.

Component	Final configuration
CNN encoder	<code>captioning.models.cnn.encoder.Cnn14Encoder</code> ; PANNs checkpoint <code>Cnn14_mAP=0.431.pth</code> ; 2048-dimensional output; encoder frozen with batch-normalization layers kept in evaluation mode.
Recurrent encoder	Three-layer bidirectional GRU; hidden size 256 per direction; 512-dimensional output; dropout 0.5.
Transformer decoder	Two layers; model/token-embedding dimension 256; four attention heads; feed-forward dimension 1024; dropout 0.2.
Positional encoding	Fixed sinusoidal positional encoding with table length 100.
Vocabulary and target length	84-token vocabulary; special tokens include <code><pad></code> , <code><start></code> , <code><end></code> , and <code><unk></code> ; maximum target length 96 tokens.
Initialization and training seeds	The five S4 runs use training seeds 1–5; the seed affects trainable-parameter initialization, data order, dropout, and scheduled sampling, while the pretrained CNN14 encoder and fixed training files remain unchanged.
Loss	Token-masked mean label-smoothed cross-entropy; label smoothing 0.05.
Optimizer	Adam; learning rate 2×10^{-4} ; weight decay 10^{-6} ; $\beta = (0.9, 0.999)$; $\epsilon = 10^{-8}$.
Learning-rate schedule	Exponential decay applied per iteration; 2,500 total iterations; 500 warm-up iterations; final learning rate 5×10^{-7} .
Training	Batch size 32; 20 epochs; gradient-norm clipping at 1.0; SpecAugment disabled.
Scheduled sampling	Enabled; linearly changed from 1.0 to 0.7 over training.
Checkpoint selection	Checkpoint with the minimum validation loss.
Main S4 inference	Beam search with beam size 3; maximum generation length 96; no length penalty.
Serialization-ablation inference	Greedy decoding (beam size 1), used for the matched five-dimensional versus Flat-22 comparison and kept distinct from the beam-decoded main S4 result.

9 Qwen3-Omni prompts and demonstration formatting

Table 14: Common Qwen3-Omni inference configuration used by the Q0–Q3 implementation.

Item	Configuration
Model	<code>Qwen3-Omni-30B-A3B-Instruct</code> .
Processor	<code>Qwen30mniMoeProcessor</code> ; <code>apply_chat_template(..., add_generation_prompt=True, tokenize=False)</code> .
Sampling / decoding	Deterministic beam search with <code>do_sample=False</code> and beam size 3.
Generation length	Maximum generation length 96; length penalty 1.0.
Sampling controls	Sampling is disabled; no stochastic sampling is used.
Audio input	Audio is inserted as an audio content item after the corresponding text instruction. With audio standardization enabled, waveforms are loaded as 16-kHz mono audio, scaled by the 99.5th percentile of absolute amplitude when nonzero, and clipped to $[-1, 1]$.

9.1 Q0: label-only zero-shot

Q0 contains no system message. The test user message consists of the following text followed by the test audio:

请根据输入音频判断是否包含下列标签。只依据音频内容选择标签，不要输出解释。

候选标签：

["起音困难", "发音中断", "明显吸气声", "沙哑/粗糙/嘶哑声", "气声", "声音颤抖", "挤压", "声音无力", "辅音模糊", "元音不饱满", "鼻音过重", "鼻音不足", "重音异常", "节奏不自然", "逐字发音", "语速过快", "语速过慢", "韵律单调", "填充词过多（嗯，啊）", "叹气", "哭腔", "笑声"]

只允许输出合法 JSON，不要输出解释文字。格式必须为：

```
{
  "abnormal": true,
```

```
"labels": []
}
```

9.2 Q1 and Q2: pathology-guided five-dimensional prompt

Q1 and Q2 use the same test-time pathology-guided instruction. Q1 is zero-shot, whereas Q2 prepends three audio-annotation demonstrations.

你是一个专业的病理语音结构化标注系统。请根据输入音频完成二阶段判定，并严格输出固定 JSON schema。

阶段1: 判断音频是否存在明确、可重复感知的病理语音异常。病理语音异常指由呼吸控制、发声质量、构音清晰度、韵律节奏或额外声学事件引起的稳定可感知异常；轻微噪声、录音设备差异或不确定现象不应标为异常。

阶段2: 仅当 abnormal 为 true 时，从下列五个维度中选择具体标签；如果 abnormal 为 false，所有维度数组必须为空。

标签库:

- 呼吸: ["起音困难", "发音中断", "明显吸气声"]
- 发声: ["沙哑/粗糙/嘶哑声", "气声", "声音颤抖", "挤压", "声音无力"]
- 构音: ["辅音模糊", "元音不饱满", "鼻音过重", "鼻音不足"]
- 韵律: ["重音异常", "节奏不自然", "逐字发音", "语速过快", "语速过慢", "韵律单调", "填充词过多 (嗯, 啊)"]
- 额外声学事件: ["叹气", "哭腔", "笑声"]

只允许输出合法 JSON，不要输出解释文字。格式必须为:

```
{
  "abnormal": true,
  "呼吸": [],
  "发声": [],
  "构音": [],
  "韵律": [],
  "额外声学事件": []
}
```

For each Q2 demonstration, the user message contains the following text followed immediately by the demonstration audio:

请按照固定 schema 标注这个音频，只输出 JSON。

The corresponding assistant message is the normalized five-dimensional expert annotation serialized as JSON. After three demonstration user/assistant pairs, the final user message contains the complete Q1/Q2 prompt above followed by the test audio.

9.3 Q3: pathology-guided Flat-22 prompt

Q3 uses the same pathology guidance, five dimensions, 22-label inventory, abnormal rule, and evidence rule as Q1/Q2. Its final output instruction is replaced by:

只允许输出合法 JSON，不要输出解释文字。格式必须为:

```
{
  "abnormal": true,
  "labels": []
}
```

For each Q3 demonstration, the user message contains the following text followed by the same demonstration audio used by Q2 for that seed:

请按照 22 个候选标签标注这个音频，只输出包含 labels 的 JSON。

The assistant demonstration target is obtained by normalizing the five-dimensional annotation and concatenating the retained labels in the fixed dimension order into the single labels array. Thus Q2 and Q3 differ in target serialization and the serialization-specific demonstration instruction, while using the same pathology guidance and matched demonstration audio.

9.4 Demonstration sets

Each Q2/Q3 demonstration set contains one expert-annotated training record from each clinical population. Q2 and Q3 use the identical ordered demonstration manifest within each seed.

10 Audio preprocessing details

Corpus segmentation and denoising were performed with the openSMILE-based preprocessing workflow. The processed corpus used by the downstream systems is stored as 16-kHz mono, signed 16-bit PCM WAV audio.

Table 15: Confirmed corpus and model-specific audio preprocessing operations.

Stage	Confirmed processing
Stored corpus	16-kHz mono, signed 16-bit PCM WAV.
SCSA	Manifest WAVs are loaded in mono and conditionally resampled to 16 kHz. The final S4 configuration uses no fixed-duration padding or cropping.
Qwen3-Omni	For the retained Q0–Q2 runs, audio is loaded as 16-kHz mono waveform, divided by the 99.5th percentile of absolute amplitude when nonzero, and clipped to $[-1, 1]$. No additional trimming, padding, cropping, or denoising is applied in this standardization step.
Weak-label voice features	Audio is loaded as 16-kHz mono waveform and unit-peak normalized. Voice-feature extraction uses activity intervals obtained with <code>librosa.effects.split</code> at <code>top_db=25</code> , <code>frame_length=2048</code> , and <code>hop_length=512</code> .
AST	Audio is resampled to 16 kHz when needed; multichannel input is averaged to mono before feature extraction.
PANNs	Audio is loaded and resampled in memory to 32-kHz mono immediately before Cnn14 inference. The Cnn14 front end uses a 1024-sample window, 320-sample hop, 64 mel bins, and a 50–14,000 Hz frequency range.
Articulation analysis	Audio is loaded at 16 kHz and trimmed with <code>librosa.effects.trim(top_db=20)</code> ; clips shorter than 0.5 s after trimming are not analyzed by this module.
Glottal / break analysis	Glottal-attack analysis uses 16-kHz audio with unit-peak normalization. The Silero-based phonation-break detector reads audio at 16 kHz with minimum silence and speech durations of 30 ms and 50 ms, respectively.

11 JSON parsing, normalization, and scanning repair

Both systems apply deterministic pre-parsing before JSON validation. The validity and repair metrics are defined at fixed stages of this pipeline so that routine schema normalization is not counted as repair.

Table 16: Confirmed JSON parsing and schema-recovery procedures.

System	Pre-parsing and normalization	Scanning-repair fallback
SCSA	Decoded text is stripped; <code><unk></code> , <code><pad></code> , <code><start></code> , and <code><end></code> are removed; whitespace is collapsed; and two fixed regular-expression corrections for known formatting errors are applied before <code>json.loads</code> . A parseable dictionary is canonicalized by inserting the five dimensions, converting permitted scalar labels to one-element lists, exact-filtering the allowed vocabulary, and removing duplicates.	The fallback is entered when JSON parsing fails or the parsed root is not a dictionary. For each dimension, the procedure scans the malformed text for the quoted dimension field and its following list region, extracts quoted strings, retains only exact allowed labels for that dimension, and removes duplicates. No synonym expansion, fuzzy matching, or label inference is used. A complete five-dimension object is constructed directly; there is no second <code>json.loads</code> success test.
Qwen3-Omni	The inference path first selects text between the first <code>{</code> and last <code>}</code> when that candidate is parseable; otherwise the raw decoded text is retained. The parser again selects the brace-delimited candidate when available and applies <code>json.loads</code> . Parseable dictionaries are normalized by unwrapping the expected object when needed, mapping Flat-22 labels to dimensions, converting permitted scalar fields to lists, exact-filtering the vocabulary, and removing duplicates.	The fallback is entered on JSON parse failure or a non-dictionary parsed root. For each dimension, the procedure scans the corresponding list region and retains exact occurrences of allowed labels. If a dimension region cannot be captured, it searches the whole candidate text for exact allowed-label strings. No synonym or fuzzy matching is used. The recovered object is then passed through the common schema normalizer.

Validity and repair metrics. Pre-parsed JSON Valid Rate is measured after the model-specific deterministic pre-parsing described above and before schema normalization or scanning repair. Schema Repair Rate is the fraction of all evaluated outputs that enter the scanning-repair fallback:

$$\text{Schema Repair Rate} = \frac{\#\{\text{outputs entering scanning repair}\}}{N}.$$